On the Power of Mediation in Games

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Bigger agenda

Motivation

- recommendation systems and delegation to AI agents create new channels for coordination

General properties of solution concepts allowing coordination (CE, CCE, BCE, etc)

- When does coordination expand what a designer can achieve?
- Which interactions are robust to coordination by potentially malicious designers?

Related work

- *Extreme Equilibria* with Kirill Rudov and Leeat Yariv
- *Robustness to Delegation in Games* with Ben Wincelberg

Today: new results on the power of mediation, with connections to both papers

Today's question

- NE: independent, decentralized play
- CE (Aumann, 1974): a mediator sends players possibly correlated recommendations
- multiple NE already make mediation useful for coordination
- when is correlation only equilibrium selection, and when does it do more?

When is $\mathrm{conv}(\mathrm{NE}) = \mathrm{CE}$ and when are they different?

A menu of motivations

The gap $\mathrm{CE}(G) \setminus \mathrm{conv}(\mathrm{NE}(G))$ is where correlation has genuine bite:

- **Design / recommendations:** can a recommender sustain outcomes beyond decentralized play?
- **Communication:** is pre-play talk just coordination on which NE to play?
  (Bárány, 1992; Forges, 1990; Lehrer and Sorin, 1997; Ben-Porath, 1998; Gerardi, 2004)
- **Learning:** can no-regret / calibrated dynamics find decentralized prediction?
  (Foster and Vohra, 1997; Fudenberg and Levine, 1999; Hart and Mas-Colell, 2000)
- **Computation:** when does a tractable LP describe a decentralized prediction?
  (Papadimitriou and Roughgarden, 2008)
- **Decision theory:** when does Bayesian rationality collapse to Nash logic?
  (Aumann, 1987)

What is known

CE can be strictly larger than $\mathrm{conv}(\mathrm{NE})$:

- the original 2×2 example (Aumann, 1974); all non-degenerate 2×2 games with two pure NE (Calvó-Armengol, 2006); mediation value in 2×2 games (Ashlagi, Monderer, and Tennenholtz, 2008)

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Extremality of a NE inside CE:

- always extreme in two-player games (Cripps, 1995; Evangelista and Raghavan, 1996; Canovas, Hansen, and Jaumard, 1999); non-extreme iff 3+ players mix (Rudov, Sandomirskiy, and Yariv, 2025)

Outline

1. Model: games, CE, NE, regularity
2. Characterization of games with $\mathrm{CE}(G) \setminus \mathrm{conv}(\mathrm{NE}(G)) \neq \emptyset$
3. How to construct points outside the hull
   - Tools: extremality, complex-valued Nash equilibria, and index theory
4. Games with a unique CE and CCE

Model

Games, correlated and NE

- finite normal-form games $G = (A = A_1 \times \ldots A_n,\ u \colon A \to \mathbb{R}^n)$

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Correlated equilibrium (Aumann, 1974)

$\mu \in \Delta(A)$ is a CE if every recommendation is self-enforcing: for all i and all $a_i, a_i' \in A_i$

$$\sum_{a_{-i}} \mu(a_i, a_{-i}) \big[u_i(a_i, a_{-i}) - u_i(a_i', a_{-i}) \big] \ \geq\ 0$$

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- $\mathrm{CE}(G)$ is a polytope: finitely many linear inequalities on $\Delta(A)$
- an NE is a CE that is a product: $\mu = \mu_1 \times \cdots \times \mu_n$
- $\mathrm{conv}(\mathrm{NE})$ = what a mediator reaches by only coordinating which NE to play

We need a non-degeneracy assumption

- pathological games blur the question
- if $u_i \equiv 0$ for all i, then $\mathrm{CE} = \mathrm{conv}(\mathrm{NE}) = \Delta(A)$
- correlation adds nothing for the wrong reason: there are too many NE

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G is regular if it has finitely many NE and each is

- quasi-strict: incentive constraints are slack outside of the support
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- non-degenerate: the Jacobian of the indifference conditions on the support is invertible

- each NE is then stable: it persists, and moves smoothly, under small payoff perturbations
- Harsanyi (1973): all games except a measure-zero set are regular
- often quasi-strictness will be enough

Characterization

Main result: a sharp dichotomy

Theorem

Let G be regular. Then exactly one of the following holds:

$$\underbrace{\mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE}) \neq \emptyset}_{\text{mediation has real bite}} \qquad \text{OR} \qquad \underbrace{\mathrm{CE} = \{\text{point}\}}_{\text{mediation powerless}}$$

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Interpretation

- communication and recommendation systems expand the set of decentralized outcomes whenever coordination failure is possible
- learning dynamics and LP solving are guaranteed to reach a decentralized outcome only in maximally robust games

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- CE (mediator knows nothing) $\subseteq$ communication equilibria (players report types) $\subseteq$ BCE (mediator omnipotent)
- $\#\text{Bayes-NE} \geq 2 \implies$ ex-ante state-independent CE, communication equilibria, and BCE are all strictly bigger that $\mathrm{conv}(\text{Bayes-NE})$

How to construct points outside the hull

Where do points outside the hull come from?

Goal: regular G and CE not a single point $\Rightarrow$ $\mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE}) \neq \emptyset$

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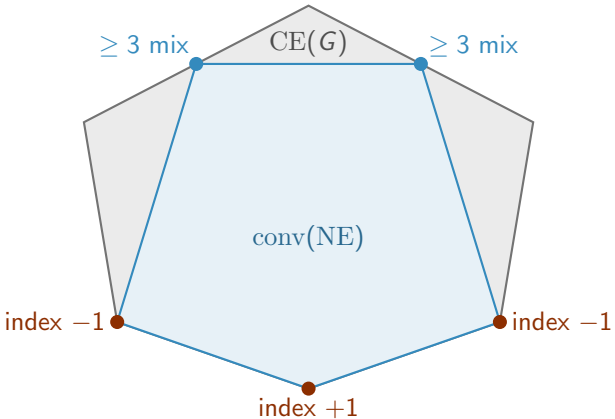
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Two regimes require different arguments:

some NE has ≥ 3 mixers:	extremality-based argument
every NE has ≤ 2 mixers:	index-theory-based argument

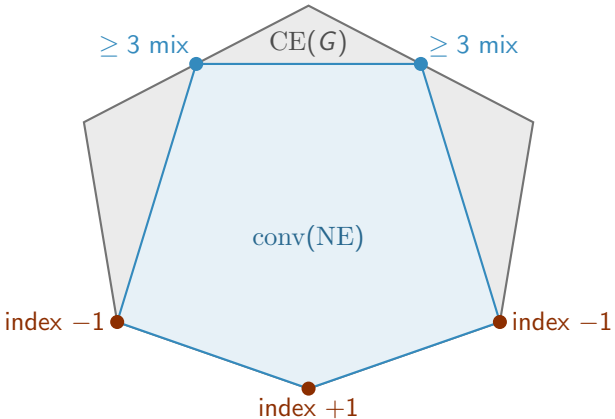
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- NE with ≥ 3 mixers sit inside faces: corners of CE escape the hull
- NE with ≤ 2 mixers sit at vertices of CE (extreme)
  - index -1 $\Rightarrow$ admit a perturbation exiting the hull

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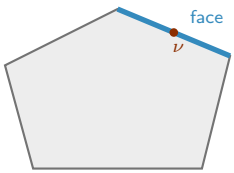
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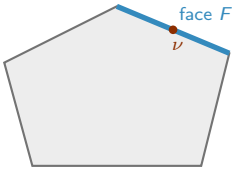


- $\Rightarrow$ those extreme points lie outside $\mathrm{conv}(\mathrm{NE})$
- in fact, all boundary points of this face are outside $\mathrm{conv}(\mathrm{NE})$

Proof sketch

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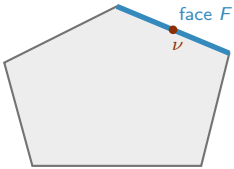
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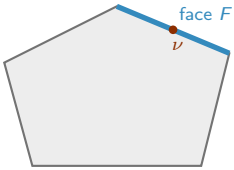
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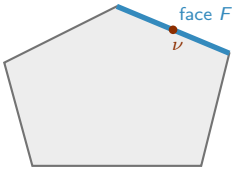
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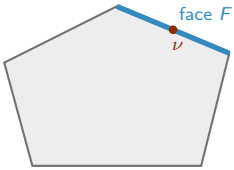
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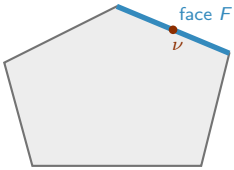
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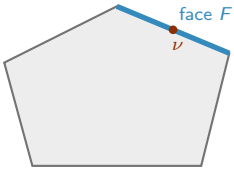
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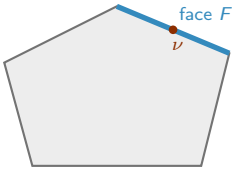
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- quasi-strictness $\Rightarrow$ all these actions are used $\Rightarrow \mathrm{supp}(e) = \mathrm{supp}(\nu)$
- so e also has ≥ 3 mixers, contradicting extremality

□

Complex-valued Nash equilibria

Fix a NE $\nu = \nu_1 \times \cdots \times \nu_n$ with ≥ 3 mixers in a regular game

Satisfies indifference conditions on $S = \mathrm{supp}(\nu)$

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Idea

- such complex solutions satisfy the real indifference conditions
- $\Rightarrow$ their real part gives a measure satisfying ICs (possibly non-product and with negative weights)
- $\Rightarrow$ a direction for an IC perturbation of the real Nash

Real perturbations from complex equilibria

How to construct

Put $h = \nu' - \nu'(A)\nu$ ($\Rightarrow$ total weight 0 and satisfies the IC). Define

$$\mu_\varepsilon = \nu + \varepsilon \operatorname{Re} h$$

Then μ_ε has mass 1 and, by quasi-strictness, is a CE for small $|\varepsilon|$

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Idea of the proof

- the boundary of the minimal CE-face carrying ν is outside the hull
- root counts for generic systems guarantee another complex solution

  Bernstein–Kushnirenko–Khovanskii theorem (Kaveh and Khovanskii, 2018), McKelvey and McLennan (1997)

Details

A toy example: four participants

A participation-coordination game with four players

- participation costs 1 and pays 8 if everyone participates
- abstaining gives payoff 0
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The fully mixed indifference conditions give a common participation probability $p = \nu_i(1)$

$$8p^3 = 1, \qquad p = \frac{1}{2},\ \frac{\omega}{2},\ \frac{\overline{\omega}}{2}, \qquad \omega = e^{2\pi \mathrm{i}/3}$$

Real solution: $\nu = (1/2, 1/2) \times \cdots \times (1/2, 1/2)$

Complex solution: $\nu' = \nu'_1 \times \cdots \times \nu'_4$, with $\nu'_i = (1 - \omega/2, \omega/2)$

The resulting mediator

Here $\nu'(A) = 1$, so $h = \nu' - \nu$ and the negative endpoint is

$$\mu = \nu - \frac{2}{21}\operatorname{Re} h, \qquad \varepsilon = -\frac{2}{21}$$

number of participants	0	1	2	3	4
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- after either recommendation, the chance all other players participate is $1/8$, so every recommendation is self-enforcing
- nonunanimous play has probability $13/14 > \underbrace{7/8}_{\text{any Nash mixture}}$

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Two ideas to the rescue

1. anti-Nash: equilibria of the auxiliary game with $u_i' = -u_i$ can help construct incentive-compatible perturbations of an NE in the original game (Cripps, 1995)
2. index theory: the construction goes through for equilibria of index -1

Crash course in index theory

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$$\mathrm{index}_u(\nu) = \operatorname{sgn} \det J(\nu), \qquad \text{where } J \text{ is the Jacobian of the indifference equations}$$

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$$u_i(b_i, \nu_{-i}) - u_i(a_i, \nu_{-i}) = 0, \qquad a_i \in \operatorname{supp}(\nu_i) \setminus \{b_i\} \quad \text{for some fixed } b_i$$

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- $J(\nu)$ is the Jacobian of all these, a $d \times d$ matrix, $d = \sum_i (m_i - 1)$, with zero diagonal blocks

$$J = \begin{pmatrix} 0 & * & * \\ * & 0 & * \\ * & * & 0 \end{pmatrix} \qquad \text{(own indifference does not depend on own mixing)}$$

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- dropping them and reversing the sign of u: $\operatorname{index}_{-u}(\nu) = (-1)^d \operatorname{index}_u(\nu)$
- $\Rightarrow$ if only two players randomize, $d = 2(m-1)$ is even, so $\operatorname{index}_{-u}(\nu) = \operatorname{index}_u(\nu)$

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Index theorem (Gül, Pearce, and Stacchetti, 1993; Ritzberger, 1994)

For a regular game, $\sum_{\nu \in \mathrm{NE}} \mathrm{index}(\nu) = +1$

one can also assign an index to equilibrium components when regularity fails, but we won't need it

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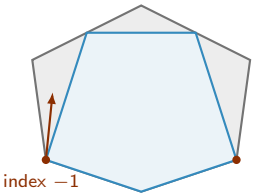
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- if ν has index -1, then ν is mixed and there is another equilibrium $\nu' \neq \nu$

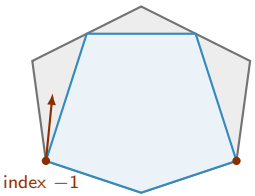
Constructing a CE outside the hull



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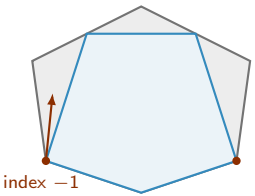
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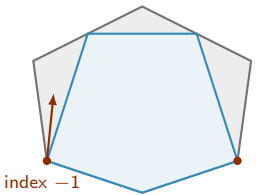
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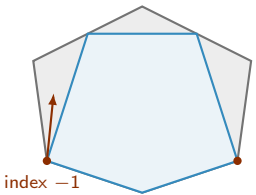
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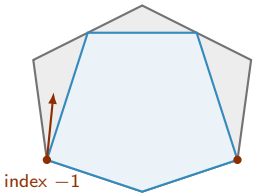
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Define,

$$\mu = \frac{\nu - \varepsilon\,\nu'}{1 - \varepsilon}$$

with $\varepsilon > 0$ small enough so that $\mu \geq 0$ and all outside-support constraints remain slack
Then $\mu \in \text{CE} \setminus \text{conv}(\text{NE})$

▸ Why outside

A toy example: subtract an anti-Nash equilibrium

A coordination game

	A	B
A	$(1,1)$	$(0,0)$
B	$(0,0)$	$(1,1)$

Consider the NE ν with
$\nu_1 = \nu_2 = (1/2, 1/2)$

$\nu = \frac{1}{4}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ has index -1

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Subtract and renormalize

$$\nu' = \delta_{(A,B)}, \qquad \varepsilon = \tfrac{1}{4}$$

$$\mu = \frac{\nu - \varepsilon\nu'}{1-\varepsilon} = \frac{1}{3}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

- $\mu \in \mathrm{CE}$
- every Nash lottery has equal off-diagonal masses
- $\mu(A,B) = 0 < \mu(B,A)$

$$\Rightarrow \quad \mu \in \mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE})$$

A unique CE

Singleton CE

skip to conclusions

Beyond a few structured classes, little is known about games with a unique CE

- concave potential games (Neyman, 1997; Ui, 2008; Cao, Tan, and Zhou, 2025), strictly socially concave games (Even-Dar, Mansour, and Nadav, 2009; Hart and Mas-Colell, 2015; Einy, Haimanko, and Lagziel, 2022), zero-sum (Forges, 1990)

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Plan

1. Characterize games with a unique CCE (Sandomirskiy and Wincelberg, 2026)
2. Outline what changes for a unique CE (work in progress)

Coarse CE

CCE (Hannan, 1957; Moulin and Vial, 1978)

$\mu \in \Delta(A)$ is a CCE if for every player i and every ex-ante deviation $a_i' \in A_i$

$$\mathbb{E}_\mu\left[u_i(a)\right] \;\geq\; \mathbb{E}_\mu\left[u_i(a_i', a_{-i})\right]$$

- the deviation is before seeing the recommendation $\Rightarrow$ CCE $\supseteq$ CE
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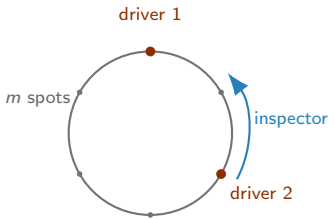
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What else?

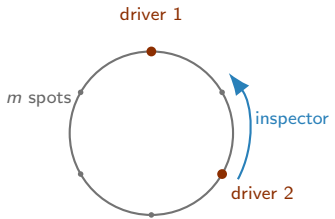
Intuition: a parking game

- 2 drivers: park illegally at one of m spots on a loop, or park legally (cost c)
- exactly one illegal $\Rightarrow$ caught for sure (ticket $t > c$)
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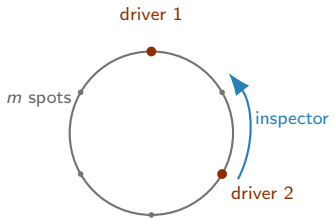
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- so legal is strictly dominant only when $\frac{t}{m} > c$
- yet legal is the unique CCE already when $\frac{t}{2} > c$
- **why:** under any joint violation at least one driver is caught with probability $\geq \frac{1}{2} \Rightarrow$ wants to deviate to legal ex-ante

Enforcement games and unique CCE

A game v is an **enforcement game** if there is $a^* \in A$ such that

$$v_i(a_i^*, a_{-i}) = 0, \quad \forall i, a_{-i}, \qquad \sum_i v_i(a) < 0 \quad \forall a \neq a^*$$

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- strategic equivalence: $u_i(a_i, a_{-i}) = \gamma_i\big(v_i(a_i, a_{-i}) + \beta_i(a_{-i})\big), \qquad \gamma_i > 0$
- **Proof idea**
  - enforcement $\Rightarrow$ unique CCE:
    $\mu(A \setminus \{a^*\}) > 0 \Rightarrow$ expected welfare $< 0 \Rightarrow \mathbb{E}_\mu[u_i] < 0$ for some $i \Rightarrow i$ would deviate to a_i^*
  - converse: LP duality

Beyond pure uniqueness

Unique CCE **can be mixed** Matching pennies: the unique NE is the unique CCE

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- in symmetric games, uniqueness $\Rightarrow$ purity
- idea: a unique CCE must be an extreme point; richer mixing destroys extremality

From CCE to CE uniqueness

Rudov, Sandomirskiy, and Yariv (2025)

A unique CE has at most two randomizing players

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- extreme-point argument: a unique CE is an extreme point of the CE polytope; ≥ 3 mixers destroy extremality
- a unique CE is quasi-strict (Viossat, 2008) $\Rightarrow$ no regularity / quasi-strictness assumption is needed

From CCE to CE uniqueness

Rudov, Sandomirskiy, and Yariv (2025)

A unique CE has at most two randomizing players

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≤ 2 mixers $\Rightarrow$ the unique CE is pure, or two players mix

we focus on the pure case

A pure unique CE (a preliminary result)

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- the gap CE vs CCE: one-shot vs multistage chains
- future: Economic interpretation? Interesting examples?

Conclusion

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Economic takeaway

- mediation, communication, and learning bring new outcomes relative to decentralized play whenever it leaves anything to coordinate on

Nerdy takeaway

- a nonlinear object, the NE, its complex counterpart, and index govern the local structure of a linear object, the CE polytope
- more surprising links likely to be found

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Thank you!

Why ≥ 3 mixers fails extremality? An example

- Game with n players, each with 2 actions

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- $\Rightarrow$ support of an extreme CE μ is bounded by $2n + 1$

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- The same argument applies to equilibria where players mix over the same number of pure strategies
- The main difficulty is handling very asymmetric equilibria

Back

Why ≥ 3 mixers rule out extremality: general case

General support bound: player i faces $|A_i|(|A_i|-1)$ incentive constraints

By Winkler (1988), every extreme CE satisfies

$$\#\,\mathrm{supp}(\mu) \leq 1 + \textstyle\sum_i |A_i|(|A_i|-1)$$

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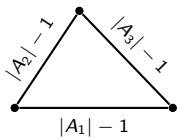
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Such examples are ruled out by the polygon inequality
If a totally mixed NE ν is an extreme CE, then

$$|A_i| - 1 \leq \sum_{j \neq i} (|A_j| - 1) \qquad \forall i$$

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- Otherwise, ν lies on a line of Nash equilibria



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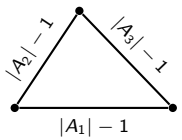
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Together: these bounds rule out extremality with ≥ 3 mixers



Triangle inequality when $n = 3$

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Back

Appendix: why another algebraic root exists

McKelvey and McLennan (1997), using Bernstein's mixed-volume theorem:
With $d_i = |\operatorname{supp}(\nu_i)| - 1$ and pure players omitted, the generic number of complex Nash equilibria is

$$B(d) = [t_1^{d_1} \cdots t_n^{d_n}] \prod_i \left(\sum_{j \neq i} t_j \right)^{d_i}$$

- the Bernstein–Kushnirenko–Khovanskii theorem counts generic roots by the normalized mixed volume of the Newton polytopes (Kaveh and Khovanskii, 2018)
- a nonsingular support Jacobian requires $d_i \leq \sum_{j \neq i} d_j$ for every i
- with 3+ mixers, these inequalities imply $B(d) \geq 2$
- binary mixers: $B(1,1,1) = 2$ and $B(1,1,1,1) = 9$

Construction

Appendix: why the anti-Nash perturbation escapes the Nash hull

Consider $\mu = \dfrac{\nu - \varepsilon\,\nu'}{1-\varepsilon}$, where ν' the anti-NE with $\mathrm{supp}(\nu')$ strictly inside $\mathrm{supp}(\nu)$

- $\Rightarrow \nu'$ omits some action $c_i \in \mathrm{supp}(\nu_i)$ for a player $i \in \{1, 2\}$
- hence the player-i marginal increases at c_i:

$$\mu_i(c_i) = \frac{\nu_i(c_i)}{1-\varepsilon} > \nu_i(c_i)$$

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- suppose for contradiction $\mu = \sum_\ell \lambda_\ell\,\nu^\ell$ with $\nu^\ell \in \mathrm{NE}$, $\lambda_\ell > 0 \Rightarrow \mathrm{supp}(\nu^\ell) \subseteq \mathrm{supp}(\nu)$
- at μ, conditional on c_i, player i is indifferent across $\mathrm{supp}(\nu_i)$
- $\Rightarrow$ the same holds under each $\nu^\ell \Rightarrow \mathrm{supp}(\nu_i^\ell) = \mathrm{supp}(\nu_i) \Rightarrow \mathrm{supp}(\nu^\ell) = \mathrm{supp}(\nu)$ for all ℓ
- in two-player games, no regular NE can share support with another NE, so every $\nu^\ell = \nu$
- then $\mu_i(c_i) = \nu_i(c_i)$, contradiction

no such decomposition exists $\Rightarrow$ $\mu \in \mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE})$

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